

LYCEUM

Multiple minds. One market.

Lyceum trades the uncertainty.

ALPACA PAPER-ONLY • AUTONOMOUS OPTIONS RESEARCH

A label is not enough for options

Most AI trading agents

BUY / SELL

Opaque conviction hides uncertainty.

Options require

- **UNCERTAINTY**
- **LIQUIDITY**
- **EXECUTION COST**

A direction can be right and the option trade can still lose.

Treat disagreement as measurable uncertainty



Each mind returns a probability distribution

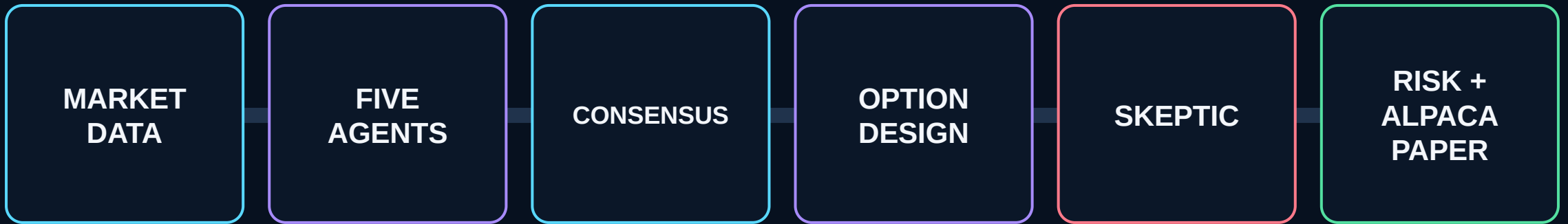
not a free-form trade command

ENTROPY

+ JENSEN-SHANNON

= explicit disagreement

AI proposes; deterministic controls decide



Models cannot submit orders or bypass the skeptic, HALT, loss limits, liquidity checks, or paper-endpoint assertion.

Evidence is chronological, cost-aware, and self-correcting

361,439

five-minute bars

666

market sessions

- CHRONOLOGICAL WALK-FORWARD
- SEP-01 INCIDENT PRESERVED
- EXECUTABLE QUOTED-SIDE P&L

No midpoint-only
profit claims.

First sealed test invalidated; repaired infrastructure awaits a clean rerun.

Execution costs dominate option economics

AUG 31 • 60-MIN REVERSION DIAGNOSTIC

+\$3.23

midpoint signal P&L



\$69.82

entry + exit crossing



Directional signals are weak.

Disagreement helps volatility forecasts modestly.

Lyceum measures the execution gap explicitly.

A research system that can say no

AI proposes.

MATH validates.

ALPACA executes.

EVERY DECISION

JOURNALED

AUDITABLE

PAPER-ONLY

Methodology first. Claims only when the evidence earns them.